

2-Fluoroethanol was easily obtained by distilling a mixture of ethylene carbonate (EC) with potassium hydrogendifluoride (KHF_2) after heated in 1-methyl-2-pyrrolidinone (NMP). The fluoroethyl methyl carbonate (FEMC) was synthesized by the reaction of 2-fluoroethanol with chloromethyl carbonate in the presence of pyridine, as shown in Fig. 1. The purification of FEMC was carried out by fractional distillation of the solution after removal of pyridine with dilute HCl solution. The formation and structure of FEMC were confirmed by GC-MS (JMS-SX102A) and ^1H , ^{13}C NMR (JEOL, JNM-LA500). The purity of FEMC was measured as 99.98% by using GC (SHIMAZU, GC-1700). Ethyl methyl carbonate (EMC), ethylene carbonate (EC) and electrolytes were used as received (battery grade). FEMC was dehydrated using purified molecular sieves (4 Å) before preparation of the solution. The preparation of the solution and the assembly of the cell was carried out in an argon-filled dry box. The apparatus and techniques for measurements of specific conductivity,⁴⁻⁶⁾ dielectric constant,⁷⁾ density⁷⁾ and viscosity⁷⁾ were reported previously. The lithium electrode cycling efficiency (charge-discharge coulombic cycling efficiency of lithium electrode) was measured with a charge/discharge unit (Ho-

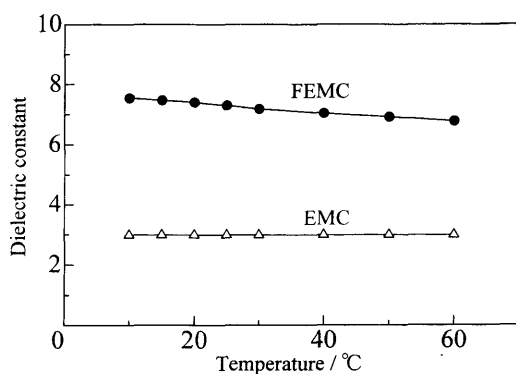


Fig. 2 Temperature dependence on dielectric constants of FEMC and EMC.

kuto Denko, HJ-201B) and estimated by a galvanostatic plating-stripping method reported by Koch and Brummer.⁸⁾ The morphology of the films formed on the Ni (working) electrode after cycling was observed by using a scanning electron microscope (SEM, JSM-5310).

3 Results and Discussion

3.1 Physical properties

Figure 2 shows the temperature dependence on the dielectric constants of FEMC and EMC. In Fig. 2, the dielectric constant of FEMC is higher than of EMC. This must be caused by the high polarity of FEMC due to the introduction of a fluorine atom with high electron withdrawing into EMC molecule. The dielectric constant of FEMC decreases linearly with increasing temperature. This may mean that the orientation of the dipoles in FEMC reduces with the rise of temperature. On the other hand, the dielectric constant of EMC with low one is almost constant at whole temperature range. Figure 3 shows the variation of densities and viscosities of FEMC and EMC with temperature. As expected from the dielectric constants in Fig. 2, the density and viscosity of FEMC are significantly higher than those of EMC. Plots of $\log \eta$ (viscosity) vs. $1/T$ for FEMC and EMC are shown in Fig. 4. These plots give the straight lines. Activation energies of viscous flow estimated from these lines are 15.9 and 10.3 kJ mol⁻¹ for FEMC and EMC, respectively. As a result, activation energy of FEMC with high viscosity becomes higher than that of EMC.

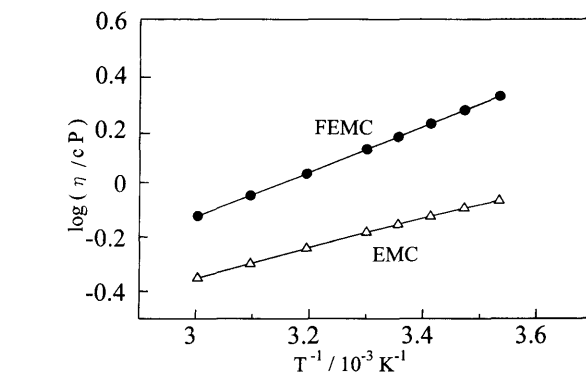
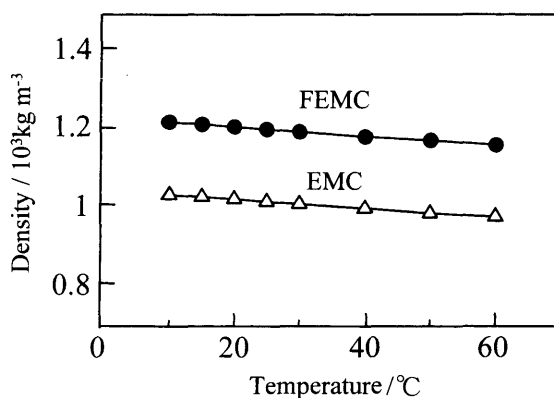


Fig. 4 Plot of $\log(\eta/\text{cP})$ vs. $1/T$ for FEMC and EMC.

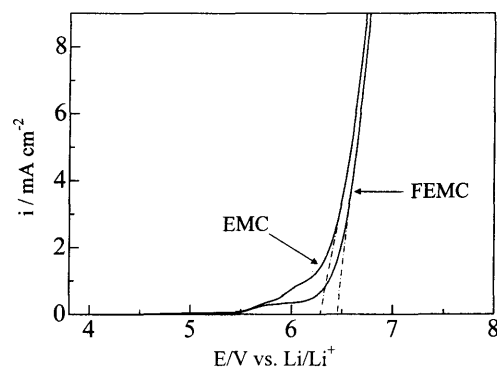


Fig. 5 i - E curves in FEMC and EMC solutions containing 1.0 mol dm⁻³ LiPF₆ using a platinum working electrode at a scan rate of 5 mV s⁻¹ at 25°C.

3.2 Electrolytic properties

Figure 5 shows i - E curves in FEMC and EMC solutions containing 1.0 mol dm⁻³ LiPF₆. In Fig. 5, oxidation durability of FEMC is slightly higher than that of EMC.

Table 1 shows the specific conductivities in FEMC and EMC solutions containing various concentrations of LiPF₆, LiClO₄ and LiBF₄. The conductivities in FEMC solutions containing 1.0 and 1.5 mol dm⁻³ LiPF₆ are lower than those in EMC solutions. This means that lower conductivities in these FEMC solutions containing LiPF₆ are dependent on the higher viscosity rather than dielectric constant of FEMC as shown in Figs. 2 and 3. At lower electrolyte concentrations such as 0.5 and 1.0 mol dm⁻³

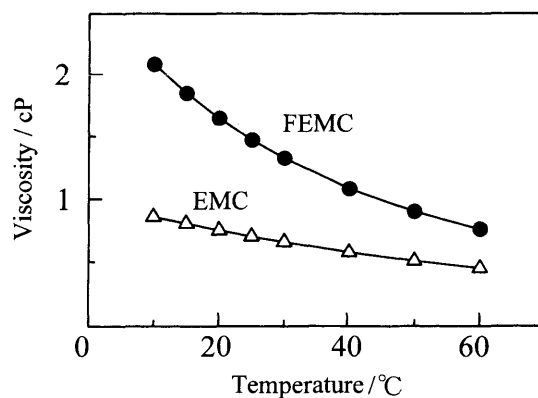


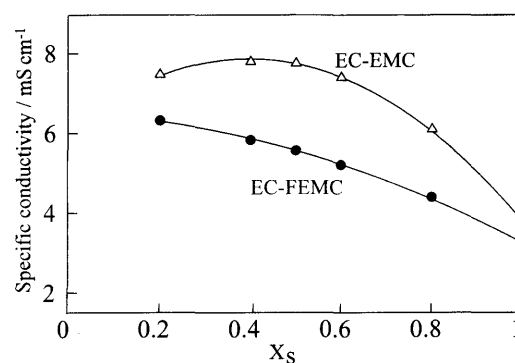
Fig. 3 Temperature dependence on densities and viscosities of FEMC and EMC.

Table 1 Specific conductivities with concentration of electrolytes in FEMC and EMC solutions at 25°C.

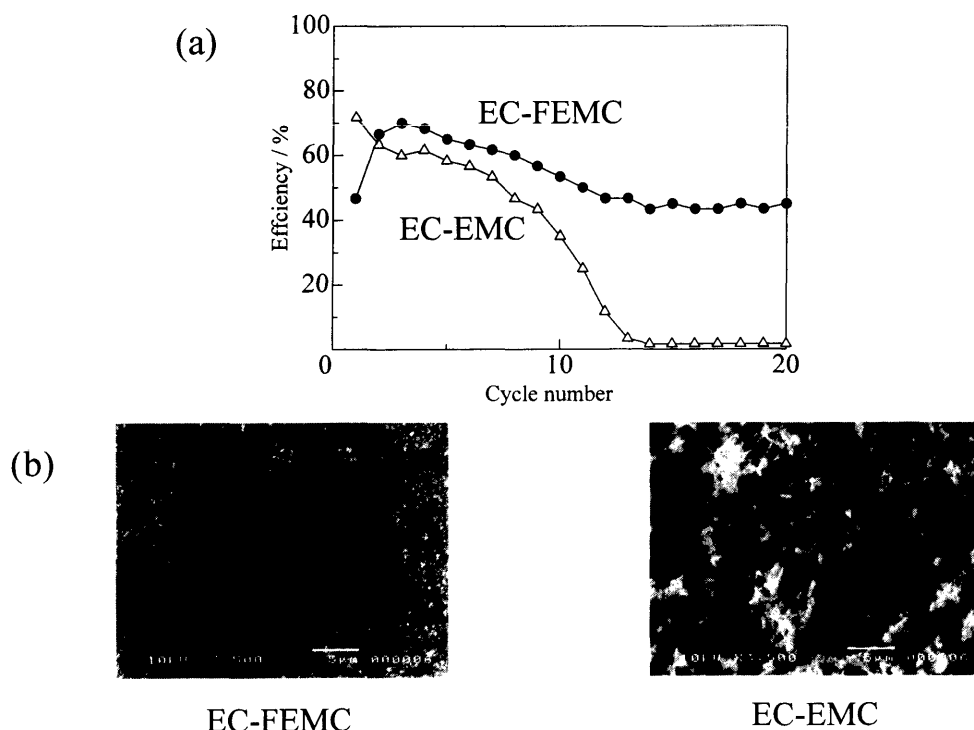
Concentration/ mol dm ⁻³	FEMC/mS cm ⁻¹			EMC/mS cm ⁻¹		
	LiPF ₆	LiClO ₄	LiBF ₄	LiPF ₆	LiClO ₄	LiBF ₄
0.5	1.98	0.67	0.21	1.24	0.25	0.01
1.0	3.22	1.38	0.53	3.76	1.20	0.16
1.5	3.20	1.65	0.81	4.96	1.99	0.58

(except LiPF₆), the conductivities in FEMC solutions are higher than those in EMC solutions. The higher conductivities in FEMC solutions may arise from the high degree of dissociation of the electrolytes due to the high dielectric constant in spite of the high viscosity of FEMC. Among these three electrolytes, the conductivities in EMC solutions containing LiBF₄ are very low. It seems that the degree of dissociation of LiBF₄ in EMC with low dielectric constant is much smaller than those of LiPF₆ and LiClO₄. In addition, the higher conductivities in FEMC solutions containing LiBF₄ compared with those in the EMC solutions seem to be dependent on the high dielectric constant of FEMC. The conductivities in mixed solvent electrolytes with FEMC and EMC were examined next.

Figure 6 shows the specific conductivities in EC-EMC and EC-FEMC binary solutions containing 1.0 mol dm⁻³ LiPF₆. The conductivity in EC-FEMC solution decreases gradually with increasing FEMC concentration, without showing the maximum conductivity like an EC-EMC solution. Figures 7 and 8 show the lithium electrode cycling efficiencies (a) in EC-FEMC and EC-EMC equimolar binary solutions containing 1.0 mol dm⁻³ LiPF₆ and LiBF₄ at 25°C, respectively, and SEM photographs (b) on

**Fig. 6** Variation of specific conductivities in EC-FEMC and EC-EMC binary solutions containing 1.0 mol dm⁻³ LiPF₆ at 25°C. X_s is the mole fraction of FEMC and EMC.

Ni electrode (working) in these solutions after discharged of the 5th and the 10th cycles. In Fig. 7(a), the efficiency in EC-EMC solution decreases rapidly with increasing cycle number. However, EC-FEMC solution shows a moderate efficiency. The surface film formed on the electrode in EC-FEMC solution compared with that in EC-EMC solution as shown in Fig. 7(b) is very homogeneous, and consists of the uniform and very small grain size. In Fig. 8(a), EC-FEMC solution shows the efficiency of more than 70% in the range of about 30th cycle. On the contrary, the efficiency in EC-EMC solution decreases appreciably after about 20th cycle. As shown in Fig. 8(b), morphology of the surface films in EC-FEMC solution analogously to that in Fig. 7(b) is very homogeneous and uniform with small grain size. Accordingly, it is thought that the high cycling efficiency in EC-FEMC solutions are related to the morphology of the

**Fig. 7** Variation of Li electrode cycling efficiencies (a) in EC-FEMC and EC-EMC equimolar binary solutions containing 1.0 mol dm⁻³ LiPF₆ at 25°C and SEM photographs (b) on Ni electrode in these solutions after discharged of the 5th cycle.

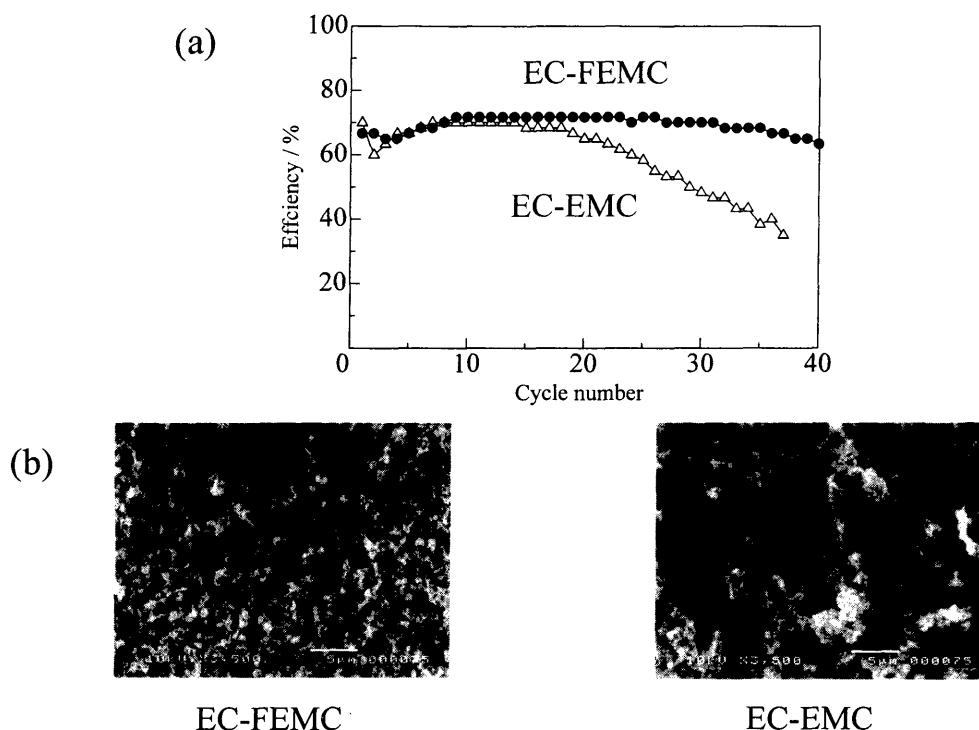


Fig. 8 Variation of Li electrode cycling efficiencies (a) in EC-FEMC and EC-EMC equimolar binary solutions containing $1.0 \text{ mol dm}^{-3} \text{ LiBF}_4$ at 25°C and SEM photographs (b) on Ni electrode in these solutions after discharged of the 10th cycle.

film on the electrode. Furthermore, it is found that EC-FEMC solutions are good electrolytes for rechargeable lithium batteries, and that the addition of FEMC to EC is very useful in an increase of the cycling efficiency.

4 Conclusions

The dielectric constant, density and viscosity of FEMC are much higher than those of EMC. The oxidative decomposition voltage of FEMC is slightly higher than that of EMC. At relatively low electrolyte concentrations, the conductivities in FEMC solutions are higher than those in EMC solutions. The specific conductivity in EC-FEMC binary solution decreases gradually with increasing FEMC concentration. The lithium electrode cycling efficiencies in EC-FEMC equimolar binary solutions containing $1.0 \text{ mol dm}^{-3} \text{ LiPF}_6$ and LiBF_4 are higher than those in EC-EMC solutions. In particular, the LiBF_4 solution shows the high efficiency of more than 70% at a higher range of cycle number. The surface of the films formed on the electrode in the EC-FEMC solutions containing

LiPF_6 and LiBF_4 is homogeneous, and consists of a uniform and small grain size.

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